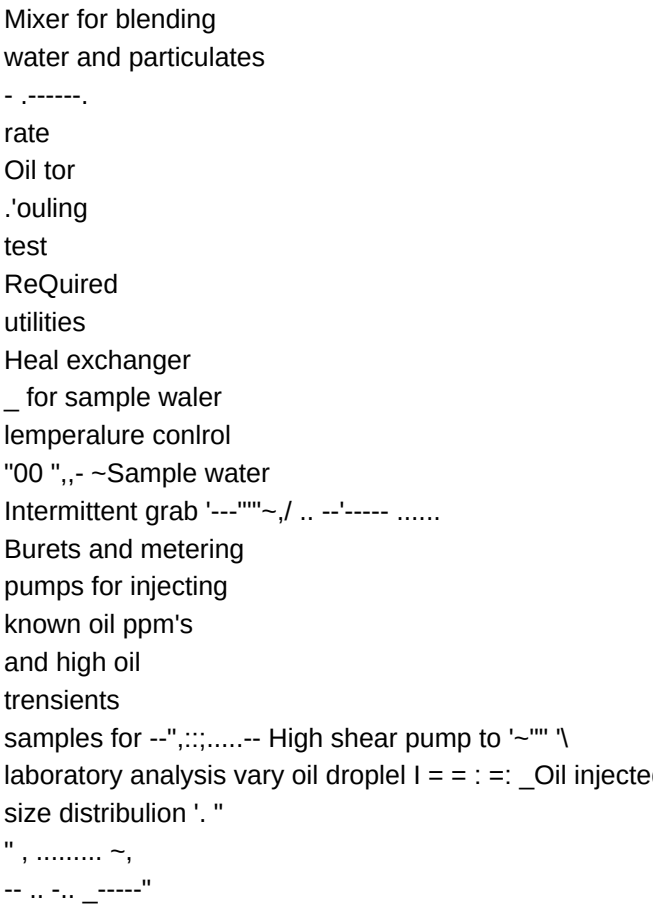
intermittent at low concentrations, the oil may be premixed with water to provide continuous flow if absolutely necessary. The oil injection point should be immediately up-stream of the oil content meter inlet to minimize time lags.

Figure 3 - Diagrammatic arrangements of test facilities



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pumps for injecting
known oil ppm's
and high oil
trensients
samples for --",:::;.....-- High shear pump to '~"" '\
laboratory analysis vary oil droplel I = = : =: _Oil injected
size distribulion '. "
" , ... ~,
-- .. -.. _-----"

2.2.5 The oil content meter will be calibrated and zeroed as per the manufacturer's instructions. It will then be tested with light distillate fuel oil of a relative density of not less than 0.83 at 15°C at the following concentrations in ppm: 0, 15, and at the full scale of the meter. Each concentration test will last for 15 minutes. Following each concentration test, the meter will be run on oil-free water for 15 minutes and the reading noted. If it proves necessary to re-zero or re-calibrate the meter during this test, this fact will be noted.

2.2.6 The oil content meter for the bilge alarms should undergo contaminant tests as follows:

the oil content meter for the bilge alarms should be run on a 10 ppm light distillate fuel oil sample, and each of the three contaminants listed below will be added to the water tank in the concentrations given. Any shift in the meter reading should be noted and recorded on the Certificate.

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SOURCE: MEPC.60(33).pdf
light distillate fuel oil sample, and each of the three contaminants listed below will be added to the water tank in the concentrations given. Any shift in the meter reading should be noted and recorded on the Certificate.
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Fresh water (if seawater is used for the test programme)
Very salt water - 6~ common salt with tap water.
Non-soluble suspended solids - about 10 ppm air cleaner test dust to the following specificqtions:
Particle si ze in mi cromet e rs Percenta ge o f to t a l weight
0- 5
5-10
10-20
20-40
40-S0
39±2
lS±3
16±3
18± 3
9±3
2.2.7 The meter should be run on a 15 ppm light distillate fuel oil sample. The water pressure or flow rate of the mixture should be adjusted from one-half normal, normal and twice normal. Any effect of these changes on the meter reading should be noted and recorded on the Certificate. This test may require modification for meters with ' flow or pressure regulators or meters

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SOURCE: MEPC.60(33).pdf
one-half normal, normal and twice normal. Any effect of these changes on the meter reading should be noted and recorded on the Certificate. This test may require modification for meters with ' flow or pressure regulators or meters
designed to disC;:,harge into an ambient pressure sump.
2.2.8 The meter should be run on a 15 ppm light distillate fuel oil sample. The water and oil injection pumps should be shut off. The meter will be left turned on with no other changes made. After eight hours, the water and oil injection pump should be turned on and set to provide the mixture of 15 ppm. The meter readings before and after each test and any damage to the meter should be noted and recorded on the Certificate. This test determines also

the proper functioning of the low flow ~hut-off and alarm.
2.2.9 If the meter requires any utilities besides electricity, it should be tested with these utilities at 110~ and 90~ of the design figures.
2.2.10 The meter should be calibrated and zeroed. A 15 ppm light distillate fuel oil sample will run through the meter for eight hours and any calibration drift noted. Following this, the meter should be run on oil-free water and

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SOURCE: MEPC.60(33).pdf
2.2.10 The meter should be calibrated and zeroed. A 15 ppm light distillate fuel oil sample will run through the meter for eight hours and any calibration drift noted. Following this, the meter should be run on oil-free water and any zero drift noted and recorded on the Certificate.
2.2.11 The response time is to . be taken for tl;1e meter to ,give an alarm ?t 15 ppm oil concentfatio ,n after the supply to the ' meter is changed fJ;'om clean water to oily water, hav.ing more than 15 ppm oil. The colotLr of the water supplied to the meter ,is not to affect the operation o,f the instrument. The meter is to be tested for colour using 2.5 ppm black ink. , The meter is to show no deviation when tested for colour using 2.5 ppm black ink.
2.2.;1.2 A specification of the instrument concerned and a diagrammatic presentation of the test arrangements should be provided and the following data should be reported, using the International Metric System of Units:
.1 types and properties of oils used in the tests;
.2 details of contaminants used"in the form, for example, of a supplier's certificate or laboratory test protocol; and
.3 results of tests and analysis of grab samples.

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SOURCE: MEPC.60(33).pdf
.1 types and properties of oils used in the tests;
.2 details of contaminants used"in the form, for example, of a supplier's certificate or laboratory test protocol; and
.3 results of tests and analysis of grab samples.
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2.2.13 The recommendations of the manufacturer of the oil content meter concerning the choice and application of cleansing agents used for cleaning purposes in machinery spaces should be recorded in the appendix to the certificate of type approval. If the manufacturer declares that the user is free in the choice and application of cleansing agents because this not affect the performance of the equipment, then this is to be recorded also. The manufacturer should include this information in the instruction manual of the oil content meter.
PART 3 - SPECIFICATIONS FOR ENVIRONMENTAL TESTING FOR TYPE APPROVAL OF POLLUTION PREVENTION EQUIPMENT
3.1 General
3.1.1 The specifications for environmental testing for type approval relate

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SOURCE: MEPC.60(33).pdf
oil content meter.
PART 3 - SPECIFICATIONS FOR ENVIRONMENTAL TESTING FOR TYPE APPROVAL OF POLLUTION PREVENTION EQUIPMENT
3.1 General
3.1.1 The specifications for environmental testing for type approval relate to the electrical and electronic sections of:
. 1 oil filtering equipment; and
.2 bilge alarms.
The equipment tested should comply with all the relevant requirements contained in section 5 of these Guidelines and Specifications.
3.2 Test specifications
3.2.1 Testing requirements
3.2.1.1 The electrical and electronic sections of the pollution control equipment in the standard production configuration should be subjected to the programme of environmental tests set out in this Specification at a laboratory approved for the purpose by the Administration or by the competent authority of the manufacturer's home country. A copy of the environmental test document, in a format similar to that specified in section 2 of part 5 of this annex, should be submitted to the Administration by the manufacturer,